



Mission Hacker Bangladesh
CYBER SECURITY TRAINING

ASSIGNMENT NO. 01

Ethical Hacking & Cybersecurity Practical

Foundation Module — Cyber Law, Lab Setup & Linux Fundamentals

BATCH – 11

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COURSE

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1. Assignment Overview

This report is my submission for the first assignment of the **Ethical Hacking & Cybersecurity Practical (Batch-11)** programme conducted by Mission Hacker Bangladesh under the guidance of mentor **Sha Jalal**. The assignment consists of four tasks covering theoretical understanding, practical laboratory setup, command-line practice and community participation. Each task is addressed in a dedicated section below.

Task	Requirement	Section	Status
01	Learn about Cybersecurity and Cyber Law and write a short note	Section 2	Completed
02	Install Kali Linux and VMware	Section 3	Completed
03	Practise all Linux commands from the MHB YouTube channel	Section 4	Completed
04	Join all communities of Mission Hackers Bangladesh	Section 5	Completed

2. Task 01 – Cybersecurity and Cyber Law: A Short Note

2.1 What is Cybersecurity

Cybersecurity is the practice of protecting computer systems, networks, programs, devices and data from unauthorised access, damage, disruption, alteration or theft. As almost every part of modern life — banking, education, healthcare, government services, business and personal communication — now depends on digital systems, protecting those systems has become as important as protecting physical property.

Cybersecurity is not a single product that can be purchased and installed. It is a continuous process built on three pillars: **people** (awareness and trained professionals), **process** (policies, standards and incident-response procedures) and **technology** (firewalls, encryption, monitoring and access control). A weakness in any one pillar can compromise the whole system, which is why attackers so often succeed through social engineering rather than through technical flaws.

2.2 The CIA Triad – the Core Principles

Every security control ultimately serves one or more of the three principles known as the **CIA Triad**. These three goals form the foundation of all cybersecurity work.

Principle	Meaning	Example of Protection
Confidentiality	Information is accessible only to those who are authorised to see it.	Encryption, strong passwords, access control lists, multi-factor authentication
Integrity	Information remains accurate, complete and unaltered by unauthorised parties.	Hashing (SHA-256), digital signatures, checksums, file-integrity monitoring
Availability	Systems and data are accessible to authorised users whenever required.	Backups, redundancy, DDoS protection, disaster-recovery planning

Two further principles are frequently added in practice: **Authentication** (proving that a user is who they claim to be) and **Non-repudiation** (ensuring a party cannot later deny an action they performed).

2.3 Major Domains of Cybersecurity

Domain	Focus Area
Network Security	Protecting network traffic and infrastructure using firewalls, IDS/IPS, VPN and network segmentation.
Application Security	Securing software against flaws such as SQL injection, XSS, CSRF and insecure authentication.
Information / Data Security	Classifying, encrypting and controlling access to data at rest and in transit.
Endpoint Security	Protecting laptops, desktops, servers and mobile devices with EDR, anti-malware and patching.
Cloud Security	Securing cloud workloads, identity and access management, and shared-responsibility configuration.
Operational Security (OpSec)	Handling of permissions, credentials and day-to-day procedures that expose or protect assets.
Digital Forensics & Incident Response	Detecting, containing, investigating and recovering from security incidents; preserving evidence.
Business Continuity / DR	Ensuring the organisation can keep operating during and after a disruptive event.

2.4 Common Cyber Threats and Attacks

Attack / Threat	Description
Phishing	Fraudulent emails, SMS or web pages that trick a victim into revealing credentials or financial data. Still the single most common entry point for breaches.
Malware	Malicious software including viruses, worms, trojans, spyware, keyloggers and rootkits.
Ransomware	Malware that encrypts the victim's data and demands payment for the decryption key.

Attack / Threat	Description
Social Engineering	Manipulating human trust — pretexting, baiting, vishing, tailgating — to bypass technical controls.
SQL Injection	Injecting crafted SQL through user input to read, modify or destroy database contents.
Cross-Site Scripting (XSS)	Injecting malicious scripts into a trusted web page so they execute in another user's browser.
Man-in-the-Middle (MITM)	Intercepting or altering communication between two parties, often on unsecured public Wi-Fi.
DoS / DDoS	Flooding a service with traffic so that legitimate users can no longer reach it.
Brute Force & Credential Stuffing	Systematically guessing passwords, or reusing credentials leaked from other breaches.
Zero-Day Exploit	Exploitation of a vulnerability before the vendor has released a patch.
Insider Threat	Damage caused by an employee or contractor, whether malicious or simply negligent.

2.5 Classification of Hackers

Type	Description
White Hat (Ethical Hacker)	Tests systems <i>with written permission</i> to find and help fix vulnerabilities. Works within a defined legal scope. This is the role this course trains for.
Black Hat	Breaks into systems illegally for personal gain, damage or theft. Criminally liable.
Grey Hat	Finds vulnerabilities without permission but usually without malicious intent. Still legally an offence in most jurisdictions, including Bangladesh.
Script Kiddie	An unskilled attacker who runs ready-made tools and exploits written by others.
Hacktivist	Attacks systems to promote a political, social or ideological message.
State-Sponsored / APT	Well-funded groups conducting long-term espionage or sabotage on behalf of a nation state.

2.6 What is Cyber Law

Cyber law (also called IT law or digital law) is the body of legislation that governs the use of computers, the internet, digital data and electronic communication. It defines what constitutes a cyber offence, prescribes punishments, establishes investigative authority, and creates rights and obligations in the digital space.

The main objectives of cyber law are to:

- Criminalise unauthorised access, data theft, hacking and digital fraud;
- Give legal recognition to electronic records, digital signatures and electronic contracts;

- Protect personal data, privacy and intellectual property online;
- Protect critical information infrastructure such as banking, power and telecommunication systems;
- Establish procedures for investigation, digital evidence and prosecution of cybercrime.

WHY CYBER LAW MATTERS TO AN ETHICAL HACKER

The technical steps performed by a penetration tester and by a criminal attacker can be almost identical — scanning, enumeration, exploitation. **The only thing that separates a professional from a criminal is documented authorisation and an agreed scope.** Knowing the law is therefore not optional background knowledge; it is what makes the profession lawful.

2.7 Cyber Law in Bangladesh

Bangladesh has developed its cyber legislation in several stages. The table below summarises the main instruments and how they relate to one another.

Legislation	Year	Scope and Status
Information & Communication Technology (ICT) Act	2006 (amended 2009, 2013)	Bangladesh's first comprehensive ICT statute. Gave legal recognition to electronic records and digital signatures and criminalised hacking, unauthorised access and digital forgery. Its controversial speech provisions (notably s. 57) were later removed and carried into subsequent laws.
Pornography Control Act	2012	Prohibits production and distribution of pornographic material, including in digital form; frequently applied in online blackmail and revenge-content cases.
Digital Security Act (DSA)	2018	Consolidated cybercrime offences, established the Digital Security Agency and a national CERT, and covered illegal access, identity fraud, cyber-terrorism and offences against critical information infrastructure. Widely criticised on freedom-of-expression grounds; subsequently repealed.
Cyber Security Act (CSA)	2023	Enacted to replace the Digital Security Act 2018. Retained the core cybercrime offences while reducing several penalties and making some offences bailable.
Cyber Protection / Cyber Security Ordinance	2025	Promulgated by the interim government to replace the Cyber Security Act 2023, repealing several speech-related provisions while retaining offences relating to hacking, unauthorised access, digital fraud and online harassment.
Digital Commerce & consumer protection rules	various	Supplementary instruments governing e-commerce, online payment and consumer redress.

NOTE ON CURRENCY OF THE LAW

Bangladesh's cyber legislation has changed repeatedly between 2018 and 2025. Before relying on any specific section number or penalty in professional work, the current consolidated text should be verified from the official gazette or from the Bangladesh Legislative and Parliamentary Affairs Division website (bdlaws.minlaw.gov.bd).

Categories of offence recognised under Bangladeshi cyber legislation

- **Illegal access** — entering a computer, network or digital device without authorisation.
- **Damage to computer or computer system** — deleting, altering or disabling data or systems.
- **Offences against Critical Information Infrastructure** — attacks on systems declared critical by the government (banking, power, telecom); these attract the heaviest penalties.
- **Identity fraud and impersonation** — using another person's identity or credentials online.
- **Digital forgery and digital fraud** — falsifying electronic records for gain.
- **Cyber-terrorism** — attacks intended to threaten sovereignty, public safety or national integrity.
- **Publication of defamatory, obscene or offensive material** in electronic form.
- **Online harassment, cyber-bullying and blackmail**, including non-consensual image sharing.

Where to report cybercrime in Bangladesh

Authority	Role
BGD e-GOV CIRT	National Computer Incident Response Team — handles incident reporting and coordination.
Cyber Crime Investigation Division, CID	Investigates cybercrime cases nationally under Bangladesh Police.
Cyber Crime Unit, DMP (Counter Terrorism & Transnational Crime)	Handles cybercrime complaints within the Dhaka Metropolitan area.
National Emergency Service — 999	General emergency reporting line, including urgent online-crime situations.

2.8 International Legal Frameworks

Framework	Summary
Budapest Convention on Cybercrime (2001)	The first international treaty on cybercrime; harmonises offences and enables cross-border cooperation and evidence sharing.
GDPR — General Data Protection Regulation (EU, 2018)	Comprehensive data-protection law granting individuals rights over their personal data; applies to any organisation processing EU residents' data.
Computer Fraud and Abuse Act (USA, 1986)	The principal US anti-hacking statute, criminalising access to a computer without or in excess of authorisation.

Framework	Summary
Information Technology Act (India, 2000)	India's core cyber legislation covering electronic records, cyber offences and intermediary liability.
ISO/IEC 27001	International standard for Information Security Management Systems (ISMS) – a compliance benchmark rather than a law.

2.9 Legal Boundaries of an Ethical Hacker

An ethical hacker performs the same technical activities as an attacker, but under a legal framework. The following conditions must always be satisfied before any testing begins:

1. **Written authorisation.** A signed engagement letter, contract or "rules of engagement" document from the legitimate owner of the system. Verbal permission is never sufficient.
2. **Defined scope.** Exact IP addresses, domains, applications and time windows that may be tested. Anything outside the scope must not be touched.
3. **Non-Disclosure Agreement (NDA).** Any data discovered during testing is confidential and must not be copied, retained or disclosed.
4. **No harm principle.** Testing must avoid denial of service, data destruction or modification unless explicitly permitted in writing.
5. **Responsible disclosure.** Findings are reported privately to the owner with sufficient time to remediate before any public disclosure.
6. **Accurate reporting.** Full documentation of methodology, evidence, risk rating and remediation advice.

Practising on systems that are not one's own and for which no permission exists is a criminal offence. For lawful practice, ethical hackers use their own laboratory environments (virtual machines such as Metasploitable, DVWA, OWASP Juice Shop) or authorised legal platforms such as TryHackMe, Hack The Box, PortSwigger Web Security Academy and public bug-bounty programmes operating under a published safe-harbour policy.

2.10 Summary of Task 01

Cybersecurity protects the confidentiality, integrity and availability of digital assets through a combination of people, process and technology. Cyber law supplies the legal structure that defines digital offences and legitimises defensive work. For a student of ethical hacking, the two subjects are inseparable: technical capability without legal authorisation is a crime, while legal knowledge without technical skill cannot defend a system. Understanding both is the foundation on which the rest of this course is built.

3. Task 02 — Installation of Kali Linux and VMware

3.1 Why Kali Linux

Kali Linux is a Debian-based Linux distribution maintained by Offensive Security and built specifically for penetration testing, digital forensics and security auditing. It ships with several hundred pre-installed security tools — including Nmap, Metasploit Framework, Burp Suite, Wireshark, Aircrack-ng, John the Ripper, Hydra and sqlmap — which removes the need to install and configure each tool individually. It is the standard working environment for this course.

3.2 Virtualization and VMware

Virtualization allows one physical computer (the *host*) to run one or more complete operating systems (*guests*) simultaneously inside isolated software containers called virtual machines. **VMware Workstation Player** is a widely used desktop hypervisor for this purpose.

Running Kali Linux inside a virtual machine offers several advantages for a learner:

- **Isolation** — testing activity is confined to the VM and cannot damage the host operating system.
- **Snapshots** — the machine state can be saved before a risky operation and restored instantly if something breaks.
- **Coexistence** — Kali runs alongside Windows without repartitioning the disk or dual-booting.
- **Lab networking** — multiple VMs (for example Kali and a vulnerable target such as Metasploitable) can be placed on a private host-only network, allowing lawful practice against a target you own.

3.3 Method A — Kali Linux on VMware Workstation Player

The following is the standard procedure for building the virtual laboratory, as demonstrated in the class and on the Mission Hackers Bangladesh YouTube channel.

Step 1 — Verify hardware and enable virtualization

- Minimum practical requirements: 64-bit processor, 8 GB RAM (4 GB absolute minimum), 40–60 GB free disk space.
- Reboot into BIOS/UEFI and enable **Intel VT-x** or **AMD-V**. Without this, the VM will not start or will run extremely slowly.
- On Windows hosts, disabling Hyper-V / "Memory Integrity" is often required for VMware to obtain hardware acceleration.

Step 2 — Install VMware Workstation Player

- Download VMware Workstation Player from the official Broadcom/VMware website only.
- Run the installer, accept the licence, keep the default installation path and complete the setup.
- Reboot the host and launch the application, selecting the free non-commercial use option.

Step 3 — Obtain the Kali Linux image

- Download from the official source: <https://www.kali.org/get-kali/>
- Two options are available:
 - **Pre-built VMware image** (recommended) — a ready-made `.7z` archive containing a `.vmx` file. Extract it and open the `.vmx` directly in VMware; no OS installation is needed.
 - **Installer ISO** — a full `.iso` image used to create and install a new virtual machine manually.
- **Verify the SHA-256 checksum** of the download against the value published on kali.org before use. This confirms the image has not been corrupted or tampered with.

Step 4 — Create the virtual machine (ISO method)

- *Create a New Virtual Machine* → select *Installer disc image file (iso)* and browse to the Kali ISO.
- Guest operating system: **Linux** → version **Debian 12.x 64-bit**.
- Name the machine (e.g. `Kali-Linux-Batch11`) and choose a storage location on a drive with sufficient free space.
- Disk size: 60 GB, stored as a single file.
- *Customize Hardware*: Memory 4096 MB (8192 MB if the host has 16 GB), Processors 2 cores, Network Adapter set to **NAT** for internet access, or **Host-only** for an isolated lab.
- Finish and power on the virtual machine.

Step 5 — Install the Kali operating system

- Select *Graphical Install* from the boot menu.
- Choose language, location and keyboard layout.
- Set the hostname (e.g. `kali`), leave the domain blank.
- Create the user account and a strong password.
- Partitioning: *Guided* — *use entire disk* and *All files in one partition*. This affects only the virtual disk, not the host machine.
- Select the desktop environment (XFCE is the default and lightest) and the default tool collection.
- Install the GRUB boot loader to the virtual disk (`/dev/sda`).
- Complete the installation and reboot into the new system.

Step 6 — Post-installation of the guest

- Update the system: `sudo apt update && sudo apt full-upgrade -y`
- Install guest tools for clipboard sharing and automatic screen resizing: `sudo apt install -y open-vm-tools open-vm-tools-desktop`
- Reboot, then take a clean **snapshot** named "Fresh Install" so the machine can always be restored to a working state.

3.4 Method B — Native Installation (Setup Used by Me)

In addition to studying the virtual-machine procedure described above, I have installed **Kali Linux directly on my laptop as the primary operating system (bare-metal installation)**. This gives full hardware access — which is particularly useful for wireless adapter work later in the course — at the cost of the snapshot and isolation benefits that a virtual machine provides.

The native installation procedure I followed was:

1. Downloaded the official Kali Linux Installer ISO from kali.org/get-kali and verified its SHA-256 checksum.
2. Wrote the ISO to a USB flash drive to create a bootable installation medium.
3. Backed up all existing personal data before proceeding.
4. Entered the UEFI/BIOS setup, disabled Secure Boot and set the USB device as the first boot option.
5. Booted the installer and selected *Graphical Install*.
6. Configured locale, keyboard layout, hostname and user account.
7. Partitioned the internal disk and installed the base system together with the KDE Plasma desktop environment and the default Kali tool set.
8. Installed the GRUB boot loader and rebooted into the installed system.
9. Ran the first full update and configured the working environment.

3.5 Installation Verification

The table below records the actual configuration of my working Kali Linux installation, as reported by the system itself using `uname`, `lsb_release`, `lscpu`, `free -h` and `df -h`.

Property	Value on my system
Operating System	Kali GNU/Linux Rolling (2026.2)
Kernel Version	6.19.14+kali-amd64
Architecture	x86_64 (64-bit)
Desktop Environment	KDE Plasma

Property	Value on my system
Processor	Intel Core i5-8265U @ 1.60 GHz
Installed Memory	15 GiB
Root Disk Capacity	221 GB
Installation Type	Native / bare-metal installation

Commands used for verification: `uname -a` · `cat /etc/os-release` · `lscpu` · `free -h` · `df -h /` · `systemd-detect-virt`

3.6 Post-Installation Configuration

Command	Purpose
<code>sudo apt update</code>	Refresh the package index from the Kali repositories
<code>sudo apt full-upgrade -y</code>	Upgrade all installed packages to current versions
<code>sudo apt autoremove -y</code>	Remove packages no longer required
<code>sudo apt install -y kali-linux-large</code>	Install the extended Kali tool collection
<code>sudo apt install -y open-vm-tools-desktop</code>	Guest tools — required only for VMware installations
<code>passwd</code>	Set or change the user account password
<code>sudo systemctl enable ssh</code>	Enable the SSH service if remote access is required
<code>ip a / ping -c 4 8.8.8.8</code>	Confirm network interface configuration and internet connectivity

OUTCOME OF TASK 02

A fully functional Kali Linux working environment has been installed, updated and verified. The VMware Workstation Player virtual-machine procedure has also been studied in full and is documented in Section 3.3, so that an isolated snapshot-capable lab VM can be built when required for exploitation exercises later in the course.

4. Task 03 – Linux Command Practice

Following the Linux tutorial videos on the Mission Hackers Bangladesh YouTube channel, I practised the commands listed below in the Kali Linux terminal. They are grouped by function for clarity. Each command was executed and its behaviour observed on my own system.

4.1 Navigation and Filesystem

Command	Function
<code>pwd</code>	Print the full path of the current working directory
<code>ls</code>	List the contents of a directory
<code>ls -l</code>	Long listing – permissions, owner, size and modification time
<code>ls -a</code>	Include hidden files (names beginning with a dot)
<code>ls -lah</code>	Long listing, all files, human-readable sizes
<code>ls -R</code>	List directories and all their contents recursively
<code>cd /path</code>	Change to the specified directory
<code>cd ~</code>	Return to the current user's home directory
<code>cd ..</code>	Move up one level to the parent directory
<code>cd -</code>	Switch back to the previous working directory
<code>tree</code>	Display the directory structure as a tree

4.2 File and Directory Operations

Command	Function
<code>mkdir dirname</code>	Create a new directory
<code>mkdir -p a/b/c</code>	Create a full nested directory path in one step
<code>rmdir dirname</code>	Remove an empty directory
<code>touch file.txt</code>	Create an empty file, or update an existing file's timestamp
<code>cp source dest</code>	Copy a file
<code>cp -r dir1 dir2</code>	Copy a directory and everything inside it
<code>mv source dest</code>	Move a file or directory, or rename it

Command	Function
<code>rm file</code>	Delete a file
<code>rm -r dir</code>	Delete a directory recursively
<code>rm -i file</code>	Delete with a confirmation prompt (safer)
<code>ln -s target link</code>	Create a symbolic link
<code>file name</code>	Identify the actual type of a file
<code>stat file</code>	Show detailed metadata: size, inode, access and modify times
<code>du -sh dir</code>	Show the total size of a directory

CAUTION

`rm -rf` deletes permanently and without confirmation — there is no recycle bin on the Linux command line. The path must always be checked before pressing Enter, and `rm -rf /` must never be run.

4.3 Viewing and Editing File Content

Command	Function
<code>cat file</code>	Display the entire contents of a file
<code>tac file</code>	Display file contents in reverse line order
<code>less file</code>	View a large file page by page (q to quit)
<code>more file</code>	Simpler page-by-page viewer
<code>head -n 10 file</code>	Show the first 10 lines
<code>tail -n 10 file</code>	Show the last 10 lines
<code>tail -f logfile</code>	Follow a log file live as new lines are written
<code>wc -l file</code>	Count the lines in a file
<code>nano file</code>	Simple terminal text editor (Ctrl+O save, Ctrl+X exit)
<code>vim file</code>	Advanced modal editor (:wq to save and quit)
<code>echo "text" > file</code>	Write text to a file, overwriting existing content
<code>echo "text" >> file</code>	Append text to the end of a file

4.4 Searching and Filtering

Command	Function
<code>grep "word" file</code>	Search for a pattern inside a file
<code>grep -i "word" file</code>	Case-insensitive search
<code>grep -r "word" /dir</code>	Search recursively through a directory tree
<code>grep -v "word" file</code>	Show only the lines that do <i>not</i> match
<code>find /path -name "*.txt"</code>	Find files by name pattern
<code>find / -type d -name conf</code>	Find directories by name
<code>locate filename</code>	Fast filename lookup using a prebuilt index
<code>which command</code>	Show the full path of an executable in \$PATH
<code>whereis command</code>	Locate the binary, source and manual page for a command
<code>sort file / uniq</code>	Sort lines; remove or count duplicate lines
<code>cut -d ":" -f 1 /etc/passwd</code>	Extract a specific field from delimited text
<code>awk '{print \$1}' file</code>	Field-based text processing
<code>sed 's/old/new/g' file</code>	Stream editing — find and replace text

4.5 Permissions and Ownership

Command	Function
<code>chmod 755 file</code>	Set permissions numerically (owner rwx, group r-x, others r-x)
<code>chmod +x script.sh</code>	Make a script executable
<code>chmod u+w,o-r file</code>	Modify permissions symbolically
<code>chown user:group file</code>	Change the owner and group of a file
<code>chgrp group file</code>	Change only the group ownership
<code>umask</code>	Show or set the default permission mask for new files
<code>sudo command</code>	Execute a single command with administrative privileges
<code>su - user</code>	Switch to another user account

Permission values: read = 4, write = 2, execute = 1. The three digits apply to *owner*, *group* and *others* respectively — so `chmod 644` gives the owner read+write (6) and everyone else read-only (4).

4.6 User and Group Management

Command	Function
<code>whoami</code>	Display the current username
<code>id</code>	Show user ID, group ID and group memberships
<code>who / w</code>	List users currently logged in and what they are running
<code>sudo adduser name</code>	Create a new user account
<code>sudo deluser name</code>	Delete a user account
<code>sudo passwd name</code>	Set or change a user's password
<code>sudo usermod -aG sudo name</code>	Add a user to the sudo (administrator) group
<code>groups</code>	List the groups the current user belongs to
<code>cat /etc/passwd</code>	View the local user account database

4.7 Process Management

Command	Function
<code>ps aux</code>	List all running processes with owner and resource usage
<code>top</code>	Live, continuously updating process monitor
<code>htop</code>	Interactive colour process viewer (easier than top)
<code>kill PID</code>	Terminate a process by its process ID
<code>kill -9 PID</code>	Force-terminate an unresponsive process
<code>killall name</code>	Terminate all processes matching a name
<code>jobs / fg / bg</code>	Manage background and foreground jobs in the shell
<code>command &</code>	Run a command in the background
<code>systemctl status service</code>	Check the state of a system service
<code>sudo systemctl start stop restart service</code>	Control a system service

4.8 Package Management (APT)

Command	Function
<code>sudo apt update</code>	Refresh the list of available packages
<code>sudo apt upgrade</code>	Upgrade installed packages

Command	Function
sudo apt full-upgrade	Upgrade, allowing packages to be added or removed as needed
sudo apt install package	Install a package
sudo apt remove package	Remove a package, keeping its configuration files
sudo apt purge package	Remove a package together with its configuration
sudo apt autoremove	Remove orphaned dependencies
apt search keyword	Search the repositories for a package
apt show package	Display detailed information about a package
dpkg -i file.deb	Install a local .deb package file
dpkg -l	List all installed packages

4.9 Networking

Command	Function
ip a (or ifconfig)	Show network interfaces and assigned IP addresses
ip route	Display the routing table and default gateway
ping -c 4 host	Test reachability of a host
traceroute host	Show the network path taken to reach a host
netstat -tulnp / ss -tulnp	List listening ports and the processes bound to them
nslookup domain / dig domain	Query DNS records for a domain
whois domain	Retrieve domain registration information
curl -I https://site	Fetch HTTP response headers from a web server
wget URL	Download a file over HTTP/HTTPS
hostname -I	Display the machine's IP address
arp -a	Show the ARP cache of known hosts on the local network
ssh user@host	Open an encrypted remote shell session
scp file user@host:/path	Copy files securely between machines

LEGAL REMINDER

Reconnaissance and scanning commands must only be pointed at systems you own or have written permission to test. Scanning third-party hosts without authorisation is an offence under Bangladeshi cyber legislation, as discussed in Section 2.9.

4.10 System Information and Monitoring

Command	Function
<code>uname -a</code>	Show kernel name, version and machine architecture
<code>cat /etc/os-release</code>	Display distribution name and version
<code>lscpu</code>	Show processor details
<code>free -h</code>	Display memory and swap usage in readable units
<code>df -h</code>	Show disk space usage per filesystem
<code>lsblk</code>	List block devices and partitions
<code>uptime</code>	Show how long the system has been running and its load average
<code>date</code>	Display or set the system date and time
<code>history</code>	List previously executed commands
<code>dmesg</code>	Print kernel ring-buffer messages
<code>lsusb / lspci</code>	List connected USB and PCI hardware devices

4.11 Archiving and Compression

Command	Function
<code>tar -cvf archive.tar dir/</code>	Create an uncompressed tar archive
<code>tar -czvf archive.tar.gz dir/</code>	Create a gzip-compressed archive
<code>tar -xzf archive.tar.gz</code>	Extract a gzip-compressed archive
<code>zip -r archive.zip dir/</code>	Create a ZIP archive
<code>unzip archive.zip</code>	Extract a ZIP archive
<code>gzip file / gunzip file.gz</code>	Compress or decompress a single file
<code>md5sum / sha256sum file</code>	Generate a checksum to verify file integrity

4.12 Shell Operators and Keyboard Shortcuts

Operator / Shortcut	Function
(pipe)	Send the output of one command into another, e.g. <code>ls -l grep ".txt"</code>
> / >>	Redirect output to a file — overwrite / append
&& /	Run the next command only if the previous succeeded / only if it failed
2>/dev/null	Discard error messages
man command	Open the manual page for a command
command --help	Show a short usage summary
clear / Ctrl + L	Clear the terminal screen
Tab	Auto-complete a file, directory or command name
Ctrl + C	Interrupt and stop the running command
Ctrl + Z	Suspend the running command to the background
Ctrl + R	Search backwards through command history
Ctrl + D / exit	Close the terminal session

OUTCOME OF TASK 03

All of the command categories above were practised directly in the Kali Linux terminal on my own machine. I am now comfortable with navigation, file handling, permissions, process control, package management and basic networking commands, which form the working foundation for the penetration-testing tools introduced later in the course.

5. Task 04 — Joining Mission Hackers Bangladesh Communities

As required by the fourth task, I have joined the official Mission Hackers Bangladesh community channels. These platforms are used for class announcements, resource sharing, assignment submission notices and peer support among Batch-11 students.

Platform	Community	Status
Facebook Group	Mission Hackers Bangladesh — official students' group	Joined
Facebook Page	Mission Hackers Bangladesh — official page	Followed
Telegram	Mission Hackers Bangladesh channel / group	Joined
WhatsApp	Batch-11 official student group	Joined

I am following all announcements posted on these channels and will continue to participate actively throughout the course, including in discussion threads and doubt-clearing sessions with the mentor and fellow students.

6. Conclusion and Learning Outcome

Completing this first assignment has given me a clear starting point for the Ethical Hacking & Cybersecurity Practical programme. From the theoretical portion I have understood that cybersecurity rests on the CIA triad and is delivered through people, process and technology together, and that cyber law — particularly the evolving legislation in Bangladesh — is what separates a professional security tester from a criminal offender. The single most important lesson is that **written authorisation and a defined scope are mandatory before any security testing**.

On the practical side, I have set up a working Kali Linux environment, verified its configuration, and studied the VMware virtual-machine procedure required for isolated laboratory work. Practising the Linux command set has made me comfortable in the terminal, which is essential because nearly every tool used in this field is operated from the command line.

I have joined all the Mission Hackers Bangladesh community channels and am ready to proceed to the next class.

7. References

1. Mission Hacker Bangladesh — Ethical Hacking & Cybersecurity Practical (Batch-11) course portal and class recordings. <https://www.missionhacker.com/course/GrcIDZrQQKFboLbXKW5w>

2. Mission Hackers Bangladesh — official YouTube channel, Linux command tutorial series.
3. Offensive Security — *Kali Linux Official Documentation*. <https://www.kali.org/docs/>
4. Bangladesh Legislative and Parliamentary Affairs Division — *Laws of Bangladesh*. <http://bdlaws.minlaw.gov.bd/>
5. BGD e-GOV CIRT — Bangladesh Computer Incident Response Team. <https://www.cirt.gov.bd/>
6. Council of Europe — *Convention on Cybercrime (Budapest Convention)*, ETS No. 185, 2001.
7. NIST — *Framework for Improving Critical Infrastructure Cybersecurity*.
8. VMware / Broadcom — *Workstation Player Documentation*.

8. Declaration

I, **Md. Abdur Razzaque**, a student of the Ethical Hacking & Cybersecurity Practical programme, **Batch-11**, conducted by Mission Hacker Bangladesh under mentor **Sha Jalal**, hereby declare that this assignment is my own work, prepared from the Class-01 lecture, the Mission Hackers Bangladesh YouTube tutorial series and the reference sources listed in Section 7.

I further declare that I understand and accept the legal and ethical responsibilities of a security practitioner, and I undertake to apply the knowledge and tools learned in this course **only on systems that I own or for which I hold explicit written authorisation**, and never for any unlawful purpose.

Md. Abdur Razzaque
Student — Batch-11

Sha Jalal
Course Mentor

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